

Understanding Taper Length Notation

Explanation of '1.0 m/s Lateral Shift' from AGTTM Part 3, Page 84

1. The Source of Confusion

In Table 5.7 of the Austroads Guide to Temporary Traffic Management (AGTTM) Part 3, the following note appears:

"Taper lengths are based on a lateral shift taper length equal to 1.0 m/s lateral shift"

At first glance, this notation appears confusing because m/s is commonly understood as meters per second — a unit of speed. However, in this context, the notation has a completely different meaning.

2. The Actual Meaning

The notation "1.0 m/s" in this context represents a ratio, not a speed measurement. It means:

For every 1 km/h of traffic speed, provide 1 meter of taper length.

This creates a consistent relationship between vehicle speed and the distance needed for drivers to safely navigate a lateral shift. The notation could be more clearly written as "1.0 m per km/h" but the standard uses the shorthand "m/s" notation.

3. The Formula

For speeds of 66 km/h and above, the lateral shift taper length can be calculated using this simple formula:

$$\text{Lateral Shift Taper Length (m)} = \text{Speed (km/h)} \times 1.0$$

Similarly, the merge taper uses a different ratio:

$$\text{Merge Taper Length (m)} = \text{Speed (km/h)} \div 0.6$$

4. Verification from Table 5.7

The table below shows how the formula produces the recommended taper lengths for various speed ranges. The "Median Speed" represents the midpoint of each speed range.

Speed Range	Median Speed	Taper Length	Calculation
66–75 km/h	70 km/h	70 m	$70 \times 1.0 = 70 \text{ m}$
76–85 km/h	80 km/h	80 m	$80 \times 1.0 = 80 \text{ m}$
86–95 km/h	90 km/h	90 m	$90 \times 1.0 = 90 \text{ m}$
96–105 km/h	100 km/h	100 m	$100 \times 1.0 = 100 \text{ m}$
$\geq 106 \text{ km/h}$	110 km/h	110 m	$110 \times 1.0 = 110 \text{ m}$

Table 1: Lateral Shift Taper Length Verification

Merge Taper Verification:

Speed Range	Median Speed	Taper Length	Calculation
66–75 km/h	70 km/h	115 m	$70 \div 0.6 \approx 117 \text{ m}$
76–85 km/h	80 km/h	130 m	$80 \div 0.6 \approx 133 \text{ m}$
86–95 km/h	90 km/h	145 m	$90 \div 0.6 \approx 150 \text{ m}$
96–105 km/h	100 km/h	160 m	$100 \div 0.6 \approx 167 \text{ m}$
$\geq 106 \text{ km/h}$	110 km/h	180 m	$110 \div 0.6 \approx 183 \text{ m}$

Table 2: Merge Taper Length Verification

5. Why Different Ratios for Different Tapers?

The two taper types use different ratios because they require different driver responses:

Lateral Shift Taper (1.0 m per km/h):

A lateral shift simply moves traffic sideways within the same roadway width. Drivers maintain their lane position relative to other vehicles and only need to follow a gradual curve. This requires less decision-making time, so a shorter taper is acceptable.

Merge Taper (0.6 m per km/h — longer taper):

A merge requires one lane of traffic to combine with another. This involves complex driver decisions: checking mirrors, judging gaps, adjusting speed, and potentially yielding to other vehicles. The longer taper provides more time and distance for these cognitive tasks.

6. Complete Taper Length Table (AGTTM Table 5.7)

Speed (km/h)	Traffic Control Taper (m)	Lateral Shift Taper (m)	Merge Taper (m)
≤ 45	15	5	15
46 – 55	15	15	30
56 – 65	30	30	60
66 – 75*	N/A	70	115
76 – 85*	N/A	80	130
86 – 95*	N/A	90	145
96 – 105*	N/A	100	160
$\geq 106^*$	N/A	110	180

Table 3: Recommended Taper Length (AGTTM Part 3, Table 5.7)

* Taper lengths for speeds ≥ 66 km/h are based on a 3.5 m lane width, lateral shift taper = 1.0 m/s, merge taper = 0.6 m/s, and median speed for each range.

7. Practical Interpretation

Understanding what the "m/s" notation actually means provides insight into driver behaviour and safety:

At 80 km/h, a vehicle travels at approximately 22 m/s. With an 80 m lateral shift taper ($80 \text{ km/h} \times 1.0$), the driver spends approximately 3.6 seconds in the taper zone. During this time, they shift laterally 3.5 m (a standard lane width), which equates to a lateral movement rate of approximately 1 m/s. This is a comfortable, gradual shift that most drivers can handle without sudden steering inputs.

The notation therefore indirectly describes the rate of lateral movement experienced by the driver — the taper length is designed so that the lateral shift occurs at approximately 1 meter per second of sideways movement at the median speed for that range.

8. Visual Comparison of Taper Types

LATERAL SHIFT TAPER	MERGE TAPER
Simple lane shift Traffic maintains flow Minimal decision-making Formula: $\text{Speed} \times 1.0$ At 80 km/h $\rightarrow$ 80 m taper	Lane closure required Traffic must merge Complex decision-making Formula: $\text{Speed} \div 0.6$ At 80 km/h $\rightarrow$ 130 m taper

Figure 1: Comparison of Lateral Shift and Merge Tapers

9. Summary

The notation "1.0 m/s lateral shift" in AGTTM Part 3 is a ratio expressing the relationship between traffic speed and required taper length. It does NOT refer to meters per second as a speed unit. The practical formulas are:

Taper Type	Ratio Notation	Formula
Lateral Shift Taper	1.0 m/s	$\text{Length (m)} = \text{Speed (km/h)} \times 1.0$
Merge Taper	0.6 m/s	$\text{Length (m)} = \text{Speed (km/h)} \div 0.6$

Reference

Austroads (2021). Guide to Temporary Traffic Management, Part 3: Static Worksites. Table 5.7, Page 77.